```
import pandas as pd

df = pd.read_csv("/content/insurance.csv")
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
df_encoded = pd.get_dummies(df, columns=["sex", "smoker", "region"], drop_first=True)
df_encoded.head()
```

	age	bmi	children	charges	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.900	0	16884.92400	False	True	False	False	True
1	18	33.770	1	1725.55230	True	False	False	True	False
2	28	33.000	3	4449.46200	True	False	False	True	False
3	33	22.705	0	21984.47061	True	False	True	False	False
4	32	28.880	0	3866.85520	True	False	True	False	False

```
X = df_encoded[["age", "bmi", "children", "smoker_yes"]]
y = df_encoded["charges"]
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("MAE :", mae)
print("MSE :", mse)
print("R2  :", r2)
```

```
MAE : 4213.798594527248
MSE : 33981653.95019776
R2  : 0.7811147722517886
```

```
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_
})

coefficients
```

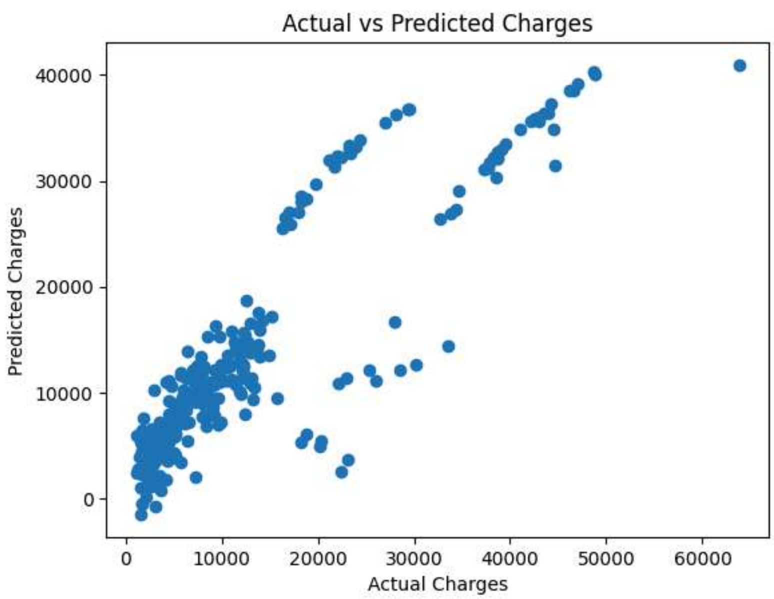
	Feature	Coefficient
0	age	257.071158
1	bmi	327.522631
2	children	427.199971
3	smoker_yes	23653.211646

```
import matplotlib.pyplot as plt

plt.figure()
plt.scatter(y_test, y_pred)

plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Actual vs Predicted Charges")

plt.show()
```

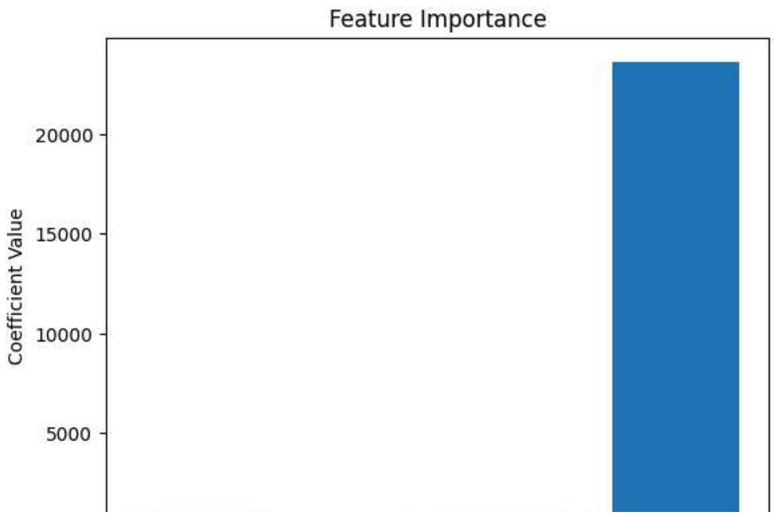



```
plt.figure()

plt.bar(coefficients["Feature"], coefficients["Coefficient"])

plt.xlabel("Features")
plt.ylabel("Coefficient Value")
plt.title("Feature Importance")

plt.show()
```


Feature Importance
Coefficient Value
20000
15000
10000
5000